Table 1: Summary Results of Simulation Study of Naive VS RRC-Corrected Effect Estimates of Cumulative Total Exposure

Generation of C	**Relative bias (%)**[a]		**Average estimated SD**[b]		**Monte-Carlo SD**[c]		**95% CI coverage**	
	Naive	RRC	Naive	RRC	Naive	RRC	Naive	RRC
Cox+Additive	Cox true $\beta = \log(1.5) \approx 0.405$, additive independent measurement error							
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	−50.81	−1.47	0.055	0.108	0.054	0.108	3.85	95.10
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	−49.67	0.77	0.056	0.111	0.056	0.112	4.95	94.60
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	−50.77	−1.41	0.056	0.112	0.056	0.112	4.00	95.30
Cox+Multiplicative	Cox true $\beta = \log(1.5) \approx 0.405$, multiplicative independent measurement error							
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	−44.92	9.90	0.065	0.130	0.074	0.147	22.90	90.30
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	−48.56	2.84	0.060	0.121	0.062	0.125	7.30	93.50
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	−49.03	1.97	0.060	0.120	0.060	0.121	6.70	95.05
ERR+Additive	ERR true $\beta \approx 0.423$, additive independent measurement error							
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	−49.71	0.01	0.068	0.135	0.069	0.140	25.00	93.75
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	−45.58	8.63	0.072	0.144	0.070	0.139	29.17	97.92
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	−45.83	8.24	0.073	0.146	0.078	0.157	29.17	96.88
ERR+Multiplicative	ERR true $\beta \approx 0.423$, multiplicative independent measurement error							
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	−52.58	−4.79	0.036	0.074	0.036	0.074	0.00	91.67
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	−53.11	−5.56	0.038	0.078	0.037	0.076	1.04	92.71
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	−53.95	−7.07	0.034	0.070	0.036	0.073	0.00	90.62

[a] $100 \times \{\text{mean}(\hat{\beta}) - \beta\}/\beta$.
[b] Mean of the estimated standard errors: model-based for Naive, sandwich estimator for RRC.
[c] Empirical standard deviation of $\hat{\beta}$ across replications.
Cox rows: main study sample size n_1=2400, validation study sample size n_2=800, simulation repetition number B=2000. ERR rows: main study sample size n_1=12000, validation study sample size n_2=4000, simulation repetition number B=96. Study duration T^*=55, cumulative incidence rate about 10%. Parameter values are in Tables 2 and 3.